

MarketTrend

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Project Description

The main purpose of our project was to use linear regression to predict the potential stock market values of six different stocks from the S&P 500 based on their performances from 2013 to 2018. We also calculated the mean squared error, the mean absolute error, and the R2 score metric values. Finally, we trained a model to predict future stock values and evaluated how accurately the model performed.

Tools and Libraries Used

We used several modules and libraries to accomplish our goals for this project:

- pandas (data manipulation and analysis)
- matplotlib and seaborn (visualizing the data into plot graphs)
- kagglehub (download and manage datasets)
- numpy (for numerical operations)
- scikit-learn
 - metrics (for model evaluation)
 - model_selection (splitting data into training/testing sets)
 - linear_model (for the linear regression model)
- os (to interact with the operating system)

Set Up Instructions

1. Download and unzip zip file
2. Ensure that all documents (including the csv) are in the same working directory
3. Run the .ipynb file in Jupyter Notebook
 - a. Run code blocks individually from top to bottom